CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 8: **Heredity** | Quick revision, one page

Basic terms

- Heredity: transmission of characters from parents to offspring. Variation: differences among individuals of a species.
- Gene: unit of DNA controlling a trait. Allele: version of a gene (T or t). Dominant: expressed in F1; recessive: expressed only when both alleles are recessive.
- Genotype: gene combination (TT, Tt, tt); phenotype: visible trait (tall/dwarf). Homozygous: same alleles; heterozygous: different alleles.
- Father and mother contribute equally to the genetic material of the child; each trait has two versions in the child.

Mendel's monohybrid cross

- Garden pea: contrasting traits (tall/dwarf, round/wrinkled seed, violet/white flower).
- P: TT $\times$ tt $\rightarrow$ F1: all Tt (tall). F1 $\times$ F1 $\rightarrow$ F2: 1 TT : 2 Tt : 1 tt.
- F2 phenotypic ratio $= 3:1$ (tall : dwarf); F2 genotypic ratio $= 1:2:1$. Tt $\times$ tt gives 1 : 1.
- Traits are inherited independently; recessive trait reappears in F2.

Sex determination in humans

- 23 pairs of chromosomes: 22 pairs autosomes + 1 pair of sex chromosomes. Female XX; male XY.
- Eggs: all carry X. Sperms: half X, half Y. XX = girl; XY = boy. Probability of boy = girl $= \frac{1}{2}$; sex decided by the father's sperm.

Variation

- Asexual reproduction: small variations from DNA copying errors. Sexual reproduction: DNA of two parents combines $\rightarrow$ more variation.
- Variations help the species survive in a changing environment.

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